

1 Solution — GIAM 6.6

Problem 4

1.1 Problem

Find a right inverse for $f(x) = |x|$.

1.2 Setup

A **right inverse** for $f : X \rightarrow Y$ is a function $g : Y \rightarrow X$ satisfying

$$f \circ g = \text{id}_Y, \quad \text{i.e.} \quad f(g(y)) = y \text{ for every } y \in Y.$$

For $f(x) = |x|$ on the natural domain $\mathbb{R}$, the image is

$$f(\mathbb{R}) = [0, \infty),$$

so any right inverse g must take values in $\mathbb{R}$ and produce outputs whose images cover $[0, \infty)$. The cleanest setup is to declare the codomain to match the image:

$$f : \mathbb{R} \rightarrow [0, \infty), \quad f(x) = |x|,$$

and look for $g : [0, \infty) \rightarrow \mathbb{R}$ with $|g(y)| = y$ for every $y \geq 0$.

1.3 A Right Inverse

The simplest choice is

$$\boxed{g : [0, \infty) \rightarrow \mathbb{R}, \quad g(y) = y.}$$

i.e. the identity on $[0, \infty)$, regarded as a function into $\mathbb{R}$.

Verification. For any $y \geq 0$,

$$f(g(y)) = |g(y)| = |y| = y,$$

so $f \circ g = \text{id}_{[0, \infty)}$. $\checkmark$

1.4 Why This Is Right but Not Two-Sided

The other composition fails. For an arbitrary $x \in \mathbb{R}$,

$$g(f(x)) = g(|x|) = |x|,$$

which equals x only when $x \geq 0$; for $x < 0$, $g(f(x)) = -x \neq x$.

So g is a right inverse but not a left inverse, and f is therefore not invertible in the two-sided sense — consistent with the general fact that

- a function has a *right* inverse iff it is **surjective**,
- a function has a *left* inverse iff it is **injective**,
- a function has a *two-sided* inverse iff it is bijective.

The map $f(x) = |x| : \mathbb{R} \rightarrow [0, \infty)$ is surjective (every non-negative real is the absolute value of itself) but *not* injective (since $f(-1) = f(1) = 1$). Right inverses exist; a left inverse does not.

1.5 Remark — Many Right Inverses

The right inverse is not unique. *Any* function $g : [0, \infty) \rightarrow \mathbb{R}$ satisfying

$$|g(y)| = y \quad \text{for every } y \geq 0$$

is a right inverse. That condition forces $g(y) \in \{y, -y\}$ for each $y > 0$, with $g(0) = 0$ (since $\{0, -0\} = \{0\}$). Two extreme choices:

- $g(y) = y$ — the positive-branch right inverse used above.
- $g(y) = -y$ — the negative-branch right inverse; equally valid.

Mixing per point also works: any function g that picks $\pm y$ for each $y > 0$ — even a “wild” non-measurable selection — gives a valid right inverse. There are uncountably many.

This non-uniqueness is the geometric counterpart of f being two-to-one off 0 : each $y > 0$ has two preimages $\{-y, y\}$, and a right inverse is exactly a *choice* of one preimage per image point. With one image point ($y = 0$) having a unique preimage and infinitely many having two, the count of right inverses is $2^{(0, \infty)}$ — a continuum-sized family.

1.6 Connection to the Section's Theme

A two-sided inverse exists exactly when the function is bijective. For $f(x) = |x|$, two natural fixes restore some form of inversion:

- **Restrict the domain** to $[0, \infty)$; on that half f is the identity and so trivially bijective with itself as inverse. (The “principal branch” approach used for parabolas in Problem 1.)
- **Pick a right inverse** on the full image; this covers all of $[0, \infty)$ but is not unique. (This problem.)

Both are useful: the first gives an exact two-sided inverse on a smaller domain; the second covers the full image at the cost of choice and non-uniqueness.